

From Concepts to Hands-On AWS

We've learned the "Why". Now let's explore the "How".

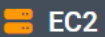
Transitioning from theory to real-world practice.

SESSION GOAL

"I know what each AWS service looks like, what it does, and how data flows through it."



S3



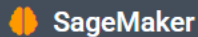
EC2



IAM



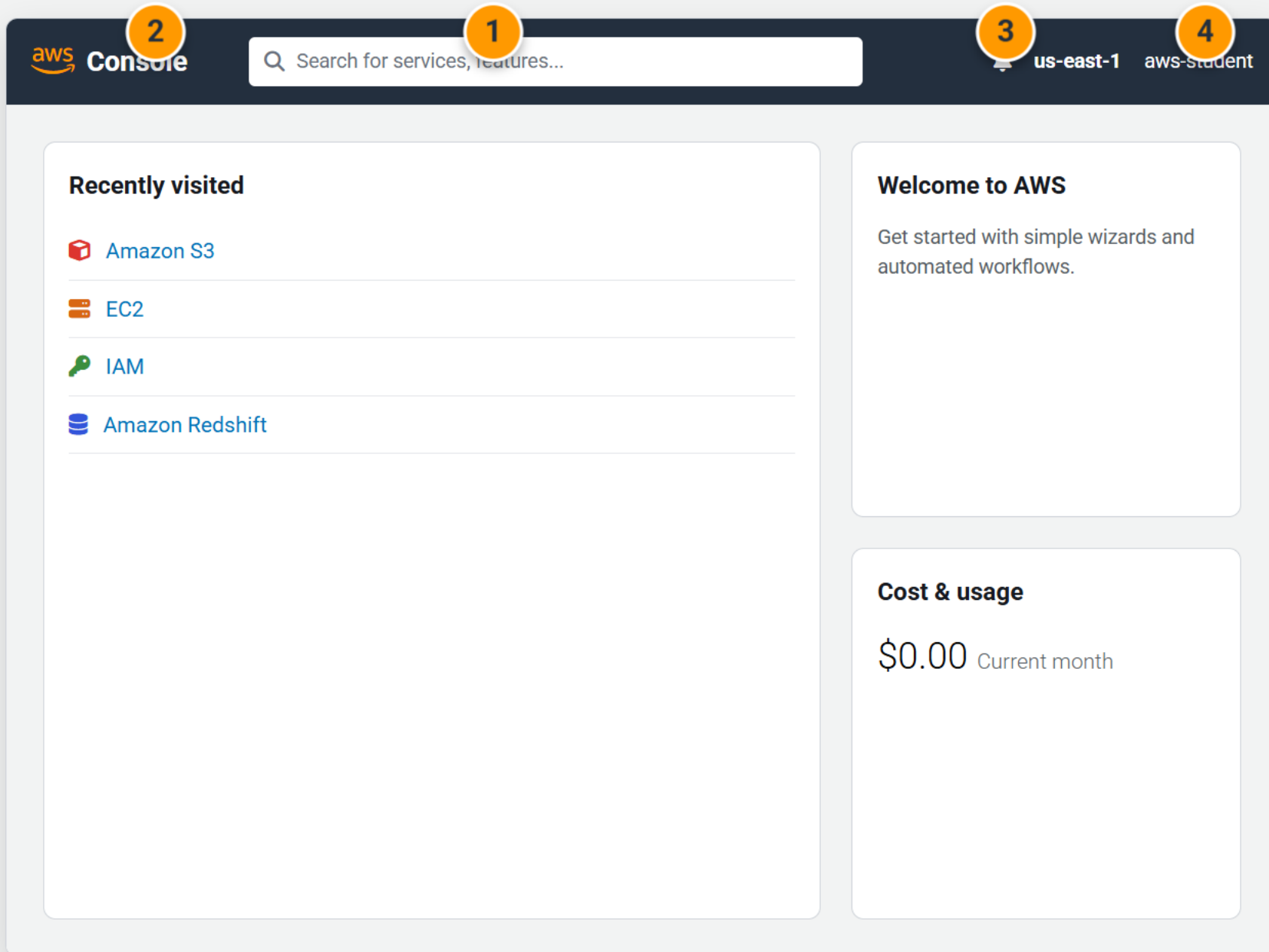
Redshift



SageMaker



Guided Lab



AWS Console Tour

Don't memorize clicks. Just know where to look.

- 1 Unified Search**
The fastest way to find anything. Type "S3", "EC2", or "Billing" here to jump straight to a service.
- 2 Services Menu**
Browse all available AWS services by category (Compute, Storage, ML, etc.).
- 3 Region Selector**
CRITICAL: Controls where your resources live physically (e.g., N. Virginia, Oregon).
- 4 Account Menu**
Access billing dashboard, security credentials, and sign out options.

BEHIND THE SCENES

The console is just a visual website that makes API calls for you. When you click buttons, AWS sends commands to its data centers.

aws Console

Search...

1

US East (N. Virginia) ▼ aws-student

Select a Region

US East (N. Virginia)	us-east-1
US East (Ohio)	us-east-2
US West (Oregon)	us-west-2
Europe (Frankfurt)	eu-central-1
Asia Pacific (Singapore)	ap-southeast-1

2

EC2 Instances

2 Running

In Region: us-east-1

Instances

0 Running

Resources are

3

Regions & Services

Location matters. Your resources live in physical data centers.

1 Region Selector

Always check this first! This dropdown controls where you are creating resources physically.

2 Physical Distance

Choose a Region close to your users (for speed) or for legal compliance (GDPR). N. Virginia (us-east-1) is the most common.

3 Resource Isolation

Resources are regional. If you create an EC2 instance in Virginia, you won't see it if you switch the menu to Oregon.

COMMON BEGINNER MISTAKE

"I created a server but I can't find it!"

👉 99% of the time, you are just looking in the wrong Region.

Create bucket

Bucket name

data-lake-project-2025

Bucket name must be unique and must not contain spaces or uppercase letters. See rules for bucket naming.

AWS Region

US East (N. Virginia) us-east-1

Object Ownership

Control ownership of objects written to this bucket.

- ☒ ACLs disabled (recommended)
- ☐ ACLs enabled

Cancel

Create bucket

1

2

What We'll Do

- 1 Enter a **globally unique name**. (e.g., add your initials or today's date).
- 2 Verify the **AWS Region**. This is where your data will physically live.
- 3 Leave all other settings as **Default** (Block Public Access is ON by default).
- 4 Click **Create bucket** at the bottom.

BEHIND THE SCENES

Why "Globally Unique"? S3 bucket names share a single global namespace (like website domains). If I create a bucket named `test`, you cannot create one named `test`—even in a different account!

 **Instructor Note:** "Ask students to raise their hands if they get a 'Bucket name already exists' error. This proves the global namespace concept!"

Amazon S3 > Buckets > data-lake-project-2025 > Upload

Upload



Drag and drop files and folders here
or click 'Add files'

Name	Size	Type
 sales_data_2024.csv	24.5 MB	text/csv

Total size: 24.5 MB

Cancel

Upload

Getting Data Into the Cloud

Follow these steps to upload your first object.

- 1 Add Files**

Click "Add files" or drag your `sales.csv` directly into this box. S3 handles files of any type (CSV, JSON, Images).
- 2 Verify Details**

Notice the file size and name. The filename becomes the "Key" (unique identifier) for that object in the bucket.
- 3 Click Upload**

Once you hit this orange button, the file travels over the internet and is stored durably in the AWS region.

BEHIND THE SCENES

This "upload" is actually a PUT Object API call. AWS S3 breaks your file into chunks, replicates it across at least 3 Availability Zones (data centers) for durability, and assigns it an HTTP URL.

 **Instructor Note:** "Point out that you don't need to provision a hard drive or format a disk. Storage scales automatically as you

S3 Hands-On: Simple Data Lake Folder Structure

Amazon S3 > Buckets > data-lake-project-2025

Objects Create folder

2 Name	Type	Last modified
 raw/	Folder	-
 curated/	Folder	-
 temp/	Folder	-
 sales_data_2024.csv	csv	Dec 24, 2025

1

What We'll Do

- 1 Inside your bucket, click **Create folder**.
Name them: raw/, curated/, and temp/.
- 2
- 3 Move your CSV file into the **raw/** folder (mock action).

BEHIND THE SCENES

Prefixes, Not Folders: S3 isn't a real file system. "Folders" are actually just prefixes in the object name (key).

Why Structure Matters? Tools like AWS Glue and Redshift use these prefixes to scan specific data sets. Separating "raw" (messy) from "curated" (clean) is standard Data Lake practice.

 **Instructor Note:** "Explain that raw/ is for untouched data straight from source, and curated/ is for cleaned data ready for analytics."



IAM User

Permanent ID Card

Represents a specific person (you) or service. Has long-term credentials like a password or access keys.

✓ Best for: Admins, Developers

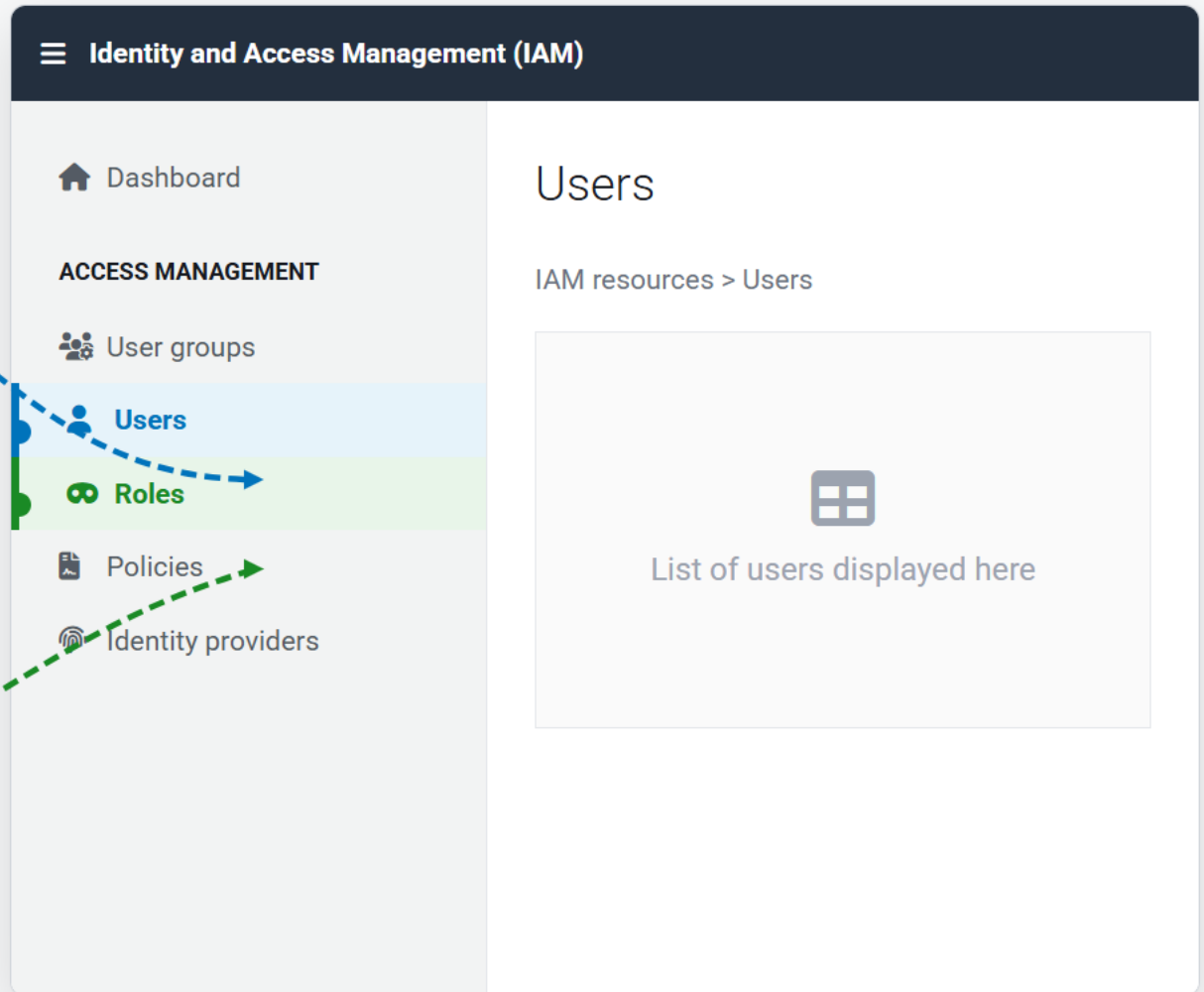


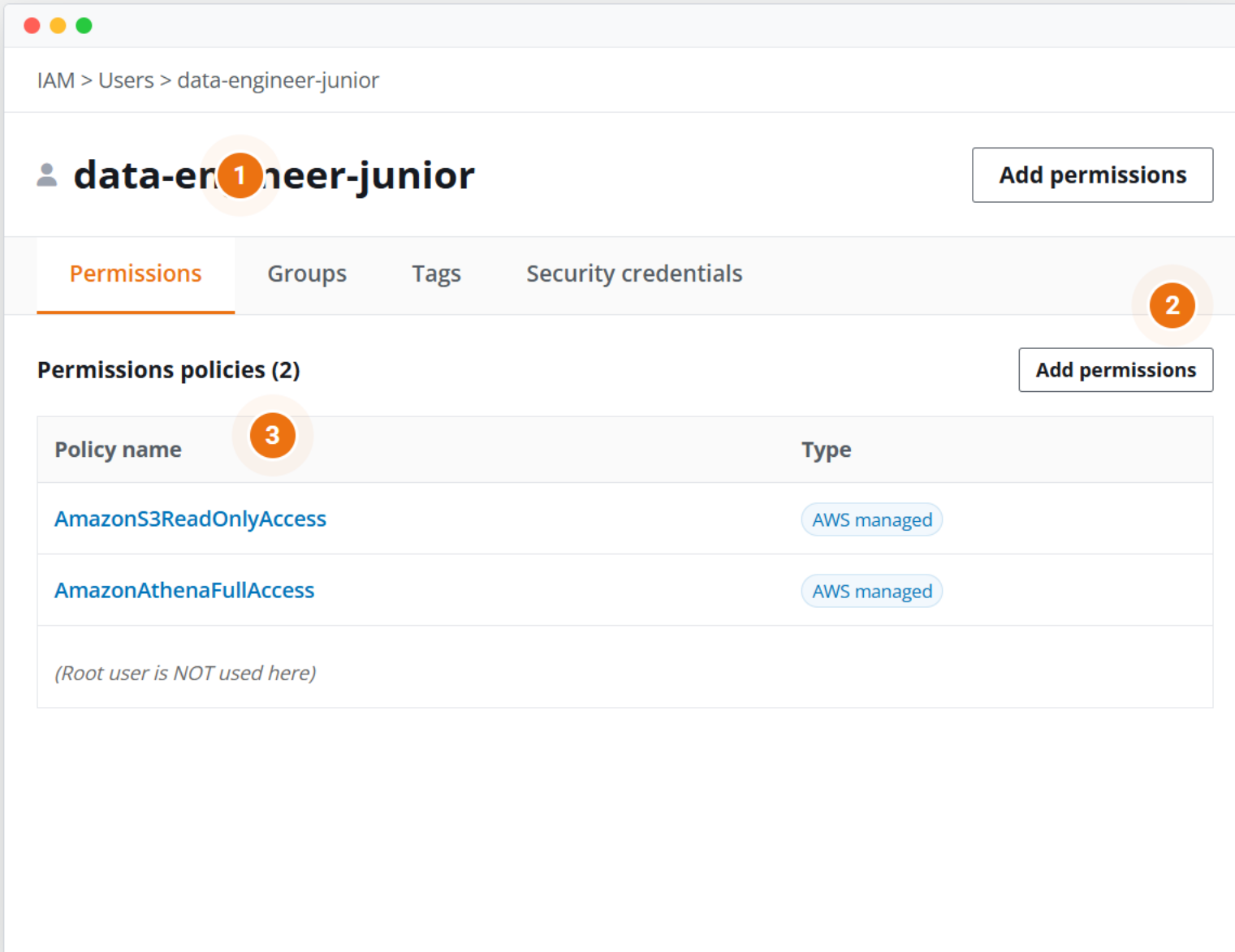
IAM Role

Temporary Hat / Uniform

Not a person. An identity that can be "assumed" temporarily. No password.

✓ Best for: EC2, Glue, Lambda





1

2

3

IAM > Users > data-engineer-junior

data-engineer-junior

Add permissions

Permissions Groups Tags Security credentials

Permissions policies (2)

Add permissions

Policy name	Type
AmazonS3ReadOnlyAccess	AWS managed
AmazonAthenaFullAccess	AWS managed

(Root user is NOT used here)

Avoiding the Root User

Create specific users and attach only necessary policies.

- 1 Permissions Tab**

For every User or Role, this tab is your source of truth. It lists exactly what actions this identity is allowed to perform.
- 2 Add Permissions**

Instead of giving "AdministratorAccess", you select specific pre-built rules (Managed Policies) like reading from S3 or querying Athena.
- 3 Least Privilege**

Notice this user has S3ReadOnlyAccess. They can see data but cannot *delete* it. This protects your data lake from accidental wiping.

SAFETY RULE #1: THE ROOT USER

The "Root User" (email login) has unlimited power. **NEVER** use it for daily work. If hacked, you lose everything. Always create an IAM User for yourself.

 **Instructor Note:** "Ask learners: 'If I wanted this user to run Glue jobs, what would I need to do here?' (Answer: Add a Glue policy)."

EC2: Launch an Instance (High-Level)

1

Name and tags

Data-Processing-Node

2

Application and OS Images (Amazon Machine Image)



Amazon Linux



Ubuntu



Windows

Amazon Linux 2023 AMI (Free tier eligible)

3

Instance type

t2.micro (1 vCPU, 1 GiB Memory)

Free tier eligible

Key pair (login)

Proceed without a key pair (Not recommended)

What We'll Do

- 1 Give the instance a **Name** (e.g., "My-First-Server").
- 2 Select an **OS Image (AMI)**. We'll stick with Amazon Linux (Free Tier).
- 3 Choose an **Instance Type**. t2.micro is small and free.
- 4 Click **Launch instance**. (We're skipping advanced networking for now!)

BEHIND THE SCENES

What is an AMI? It's the "template" for your computer. It contains the Operating System (Linux, Windows) and pre-installed software.

Instructor Note: "Point out the 'Free tier eligible' tags. This is crucial for beginners to avoid accidental costs. Explain that 't2.micro' is just a code for 'small computer'."

EC2 > Instances

Instances (1, 1)

Launch instances

Connect

Instance actions

Stop instance

Start instance

Reboot instance

Terminate instance

	Name	Instance ID	Instance State	Type	Status Check
☒	data-work	i-0c3d4e	● Running	t3.micro	✓ 2/2 checks passed
☐	web-server-dev	i-0f9e8d7c6b	■ Stopped	t3.micro	-

Details

Security

Networking

Storage

Status checks

Monitoring

✓ System status checks: passed

✓ Instance status checks: passed

3

Managing Compute Lifecycle

Understanding what "Running" means for your wallet.

- 1 Control the State**

Use the **Instance state** menu to Stop, Start, or Reboot. Always **Stop** your instance when you aren't using it to save money.
- 2 "Running" = Billing**

When the state is green ("Running"), AWS is charging you per second. When it says "Stopped," you only pay for the storage attached, not the compute.
- 3 Health Monitoring**

The bottom tabs show critical info. **Status Checks** tell you if the hardware or software is healthy. If a check fails, a reboot often fixes it.

BEHIND THE SCENES

Stop vs. Terminate: "Stopping" is like shutting down your laptop—data is saved on the hard drive (EBS). "Terminating" is like throwing the laptop in the trash—the instance and its local data are permanently deleted.

Add crawler

Crawler name

s3-raw-data-crawler

Choose a unique name to identify this crawler.

Data sources

Select the data store to crawl.

▼  s3://data-lake-project-2025

📁 raw-zone/

📁 curated-zone/

Target database

retail_db ▼

Where the metadata tables will be created.

Cancel

Create crawler

1

2

What We'll Do

- 1 Create a **Crawler** and give it a name like s3-raw-data-crawler.
- 2 Point it to your S3 bucket's **raw-zone/** folder as the data source.
- 3 Select a **Target Database** (the virtual container in the Data Catalog).

BEHIND THE SCENES

Schema Discovery:

The Crawler opens your CSV/JSON files, reads the headers and data types (String, Int), and creates a "Table Definition" in the Data Catalog.

 **Instructor Note:** "Show them the 'Tables' section after the run. They should see a table named after their folder (e.g., 'raw_zone') with columns populated automatically."

AWS Glue > ETL jobs > Visual ETL > clean_sales_data

clean_sales_data Draft

Save Run

Visual Script Job details Runs Data quality

1

S3 Bucket
raw-zone

Mapping
Rename cols

Redshift
sales_db

Node properties

Name
S3 Bucket

S3 URL
s3://my-bucket/

Data format
CSV

ETL Without Writing Code

Define how data flows and transforms visually.

- 1 Visual Graph**
Drag and drop sources (S3) and targets (Redshift). AWS Glue draws the flow for you. Each box represents a step in your data pipeline.
- 2 Tabs (Script / Details)**
Use "Job details" to set the IAM Role (permissions) and worker type. Use "Script" to see the Python code Glue generates automatically.
- 3 Run Button**
Clicking "Run" spins up serverless workers to process your data at scale. You pay only for the seconds the job runs.

BEHIND THE SCENES

Glue is essentially "Serverless Spark". When you click Run, AWS provisions a cluster, distributes your data across nodes, processes it in parallel, and then terminates the cluster.

Create cluster

Cluster configuration

Cluster identifier

A unique name for your cluster.

Node type

Choose the lowest cost node for training purposes.

Database configurations

Admin user name

Admin user password

1

2

3

What We'll Do

- 1 Set a unique **Cluster identifier**. This names your data warehouse.
- 2 Select **Node Type**. For learning, use the smallest available (like dc2.large) to minimize cost.
- 3 Create an **Admin user & Password**. You will run SQL queries later by logging in.

BEHIND THE SCENES

Leader vs. Compute Nodes:

When you click "Create", AWS launches a set of servers. One "Leader" manages connections, while "Compute" nodes store data and run queries in parallel.

⚠ Safety Tip: "Redshift can be expensive! Remind students we are creating this for the demo only and will PAUSE or DELETE it immediately after finishing."

Amazon Redshift > Query editor v2 > Editor

dev

public

sales_data

1

inventory

Query 1

2

Run

Save

```
SELECT product_id, category, amount
FROM sales_data
WHERE category = 'Electronics'
LIMIT 5;
```

3

Results (5 rows returned)

product_id	category	amount
P-1001	Electronics	\$1,299.00
P-1045	Electronics	\$499.50
P-2022	Electronics	\$89.99

4

Querying Your Data Lake

Use standard SQL to analyze data without moving it.

- 1 Select Source Table**

Locate your `sales_data` table in the sidebar. This table structure was created by the Glue Crawler we ran earlier.
- 2 Click Run**

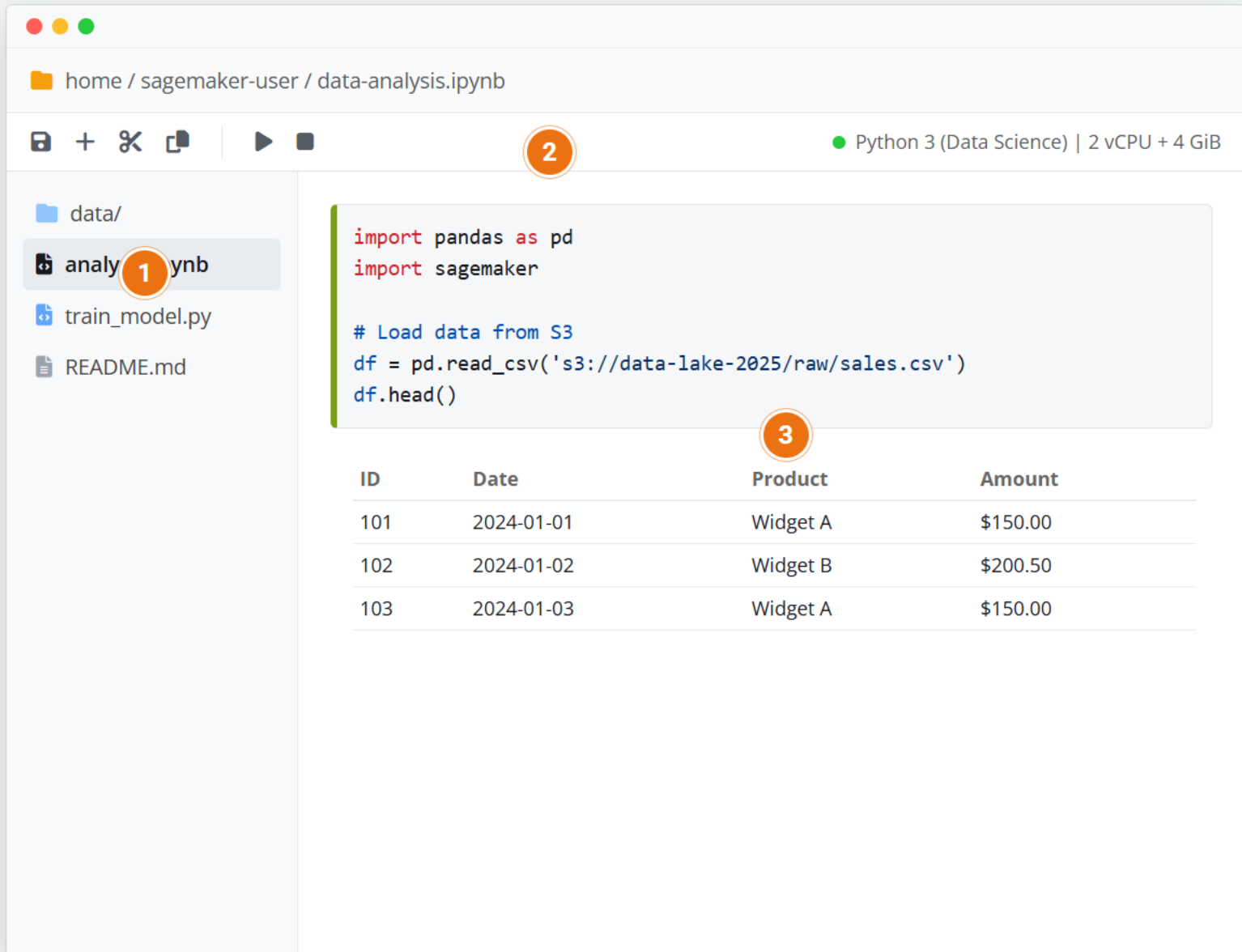
After typing your SQL query (Standard SQL), click the orange **Run** button. Redshift creates a plan to execute your request.
- 3 Write SQL**

Enter commands like `SELECT *` to view data. You don't need to know how the files are stored in S3; Redshift handles that.
- 4 View Results**

Data appears instantly in the grid below. You can now download this as a CSV or visualize it.

BEHIND THE SCENES

When you click "Run", the **Leader Node** parses your SQL and distributes the work to **Compute Nodes**. If using Redshift Spectrum, these nodes fetch data directly from S3 on demand.



The screenshot shows the SageMaker Notebook Interface. At the top, the path is `home / sagemaker-user / data-analysis.ipynb`. Below the path is a toolbar with icons for file operations and a play button (labeled 2). To the right of the toolbar, the kernel is identified as `Python 3 (Data Science) | 2 vCPU + 4 GiB`. On the left sidebar, a file browser shows a folder `data/` and files `analy1ynb` (labeled 1), `train_model.py`, and `README.md`. The main area contains a code cell (labeled 3) with the following Python code:

```
import pandas as pd
import sagemaker

# Load data from S3
df = pd.read_csv('s3://data-lake-2025/raw/sales.csv')
df.head()
```

Below the code cell, the output is displayed as a table:

ID	Date	Product	Amount
101	2024-01-01	Widget A	\$150.00
102	2024-01-02	Widget B	\$200.50
103	2024-01-03	Widget A	\$150.00

Your Cloud ML Workspace

An interactive environment to write code and visualize data.

1 File Browser

Navigate your files just like on your laptop. You can access scripts, upload datasets, or browse files synced from S3.

2 Run Button & Kernel

Click "Play" to execute code. The "Kernel" (top right) is the engine running your code—here, a Python environment pre-installed with data science libraries.

3 Notebook Cells

Write code in chunks. In this example, we're pulling data directly from our S3 bucket into a DataFrame for analysis.

BEHIND THE SCENES

Even though this looks like a webpage, it's actually running on a dedicated **EC2 instance** managed by SageMaker. When you stop working, remember to shut down the instance to stop the billing meter!



Model Training

GPU HEAVY

"Studying for the Exam"

The intense process of teaching the model. Requires massive computing power (GPUs) but runs only temporarily.

 **Cost Tip: Stop immediately after finishing!**



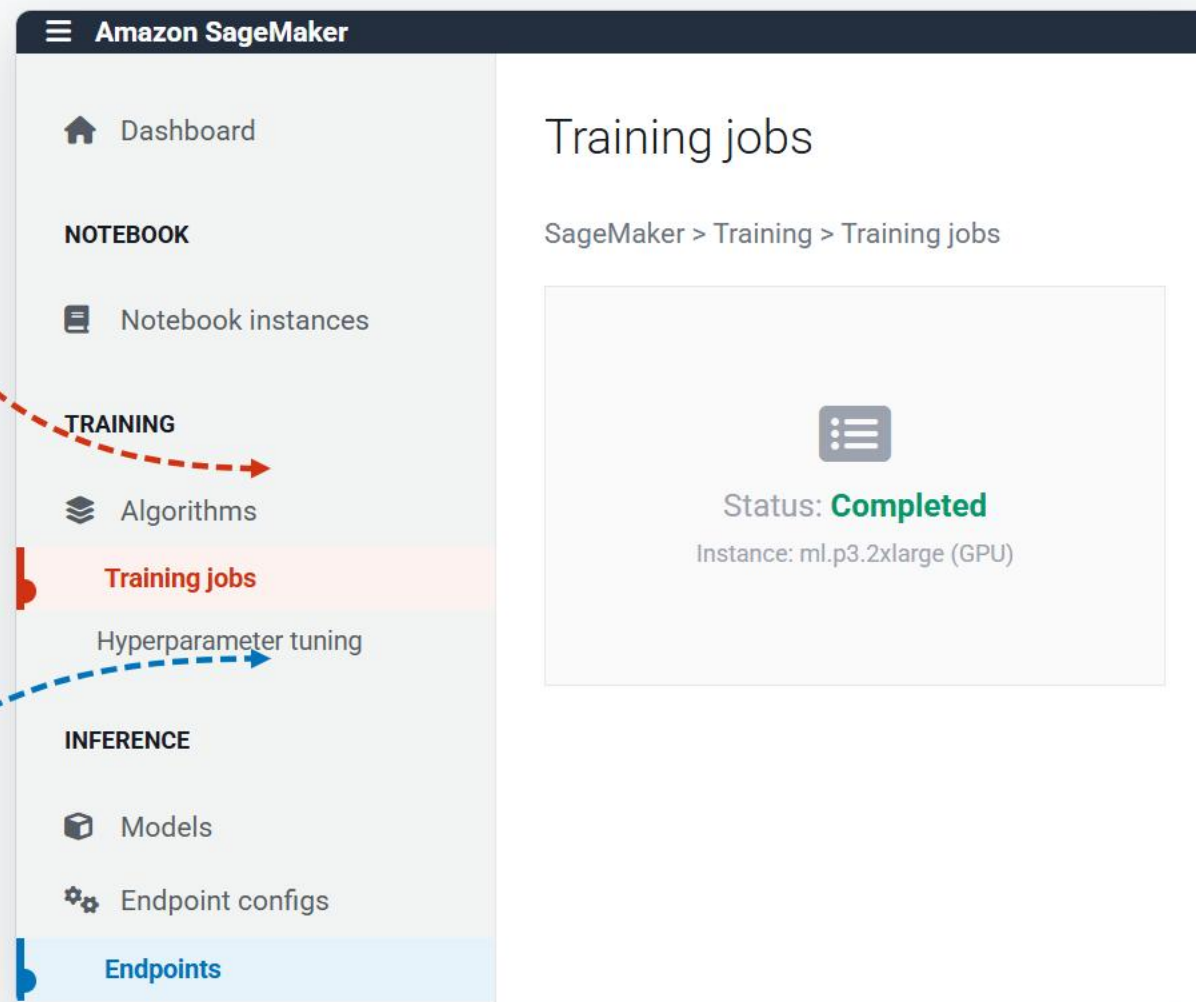
Inference (Hosting)

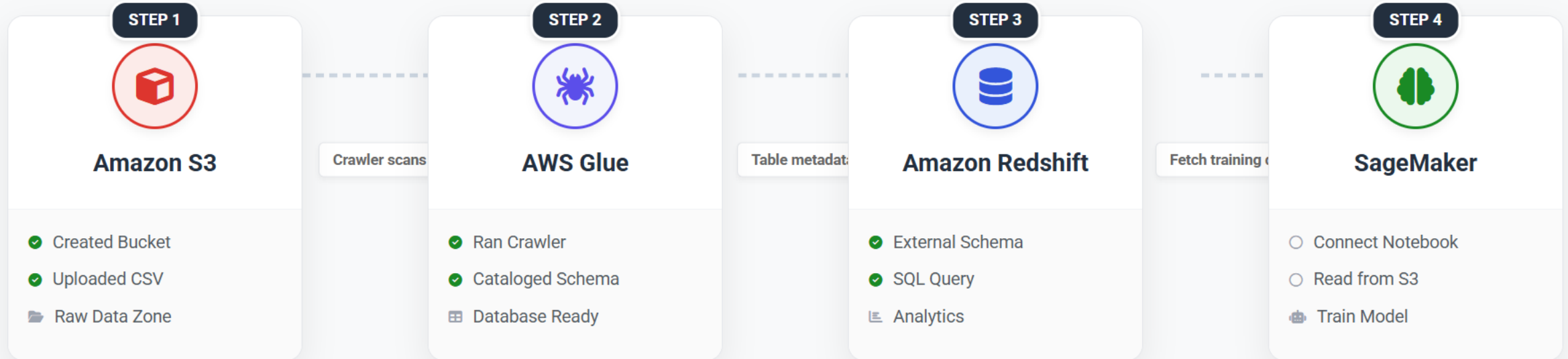
CPU / LIGHT GPU

"Using the Knowledge"

The running model that answers questions. Often runs 24/7 on smaller, cheaper instances (endpoints).

 Use: Real-time predictions or Batch transform





The "Lake House" Architecture in Action

We stored raw data cheaply in S3 (Lake), used Glue to understand its structure, and queried it immediately with Redshift (Warehouse) without moving the data. This decoupled storage from compute, saving costs and time.

STOP: The Root User Trap

Mistake: Using your root account email/password for daily work.

Why it's bad: The root user has unlimited power. If compromised, attackers can delete everything or spin up expensive resources instantly. You cannot restrict the root user.

✓ **Fix:** Create an IAM Admin User and lock the root credentials away.

CAUTION: The "Ghost" Resources

Mistake: Closing the browser tab without stopping the resource.

Reality: AWS bills for *provisioned* time, not *active* usage. An idle EC2 instance running 24/7 costs money even if you aren't logged in.

 **Region Trap:** Resources in N. Virginia are invisible if you switch the console to Oregon. Always check your region!

The "Log Off" Protocol

Before you finish your session, always check these dashboards:



EC2 Instances

State must be **Stopped** or **Terminated**. "Running" = \$\$\$



Redshift Clusters

Delete the cluster when done. Pausing helps, but snapshots store data.



SageMaker Notebooks

Must be **Stopped**. Closing the Jupyter tab does NOT stop the underlying instance.



Glue Endpoints

Check "Dev Endpoints" and ensure Crawlers are not on a repetitive schedule.



Free Tier Reality Check

The Free Tier has limits (e.g., 750 hours of t2.micro). If you launch two instances, you burn through those hours twice as fast. Always monitor **AWS Budgets**.

♥ Cloud computing is powerful. Use it responsibly!